DS-A1 | Data Provenance and Measurement Audit

Access git-hub: https://github.com/alldeolove10-SJ/2026A8000531008_DEONISIO-YOHANES-WILFRIDUS-HORNAY_Week3_DS_A1_Data-Provenance-and-Measurement-Aud

Worked submission draft - UCI Iris dataset

Source: UCI Machine Learning Repository, Iris, DOI 10.24432/C56C76

URL: https://archive.ics.uci.edu/dataset/53/iris

License: CC BY 4.0

Frozen file: data/iris_frozen.csv

SHA-256: 7e4e13e60d330d83ae5309b440eb5ececb4b779e925cc621d8f347b1e14eeca3

1. Question and measurement boundary

Question: Within this dataset, how well do petal measurements separate the three recorded iris species, and what data-quality limitations constrain that conclusion?

Element	Definition
Stakeholder	Teaching/research team demonstrating measurement-aware exploratory data science
Population	Iris flowers of the three represented species under comparable measurement procedures
Sample	150 measured flowers in the frozen copy
Unit	One flower
Target	Recorded species label
Estimand	Difference in mean petal length between species groups in this observed sample

2. Provenance lineage

Real iris flower -> physical sepal/petal measurements and species assignment -> historical dataset compilation -> UCI repository -> frozen local CSV -> pandas audit and descriptive analysis. Measurement, labeling, transcription, repository transformation, and preprocessing are all potential error points.

3. Audit results

Check	Result
Rows / columns	150 / 5
Missing values	0
Exact duplicate rows	1
Classes	setosa, versicolor, virginica
Positive numeric measurements	Yes

IQR-based anomaly flags are treated as prompts for review rather than proof of error. A later predictive model must exclude the species target from predictor features and must fit preprocessing without using test-set information.

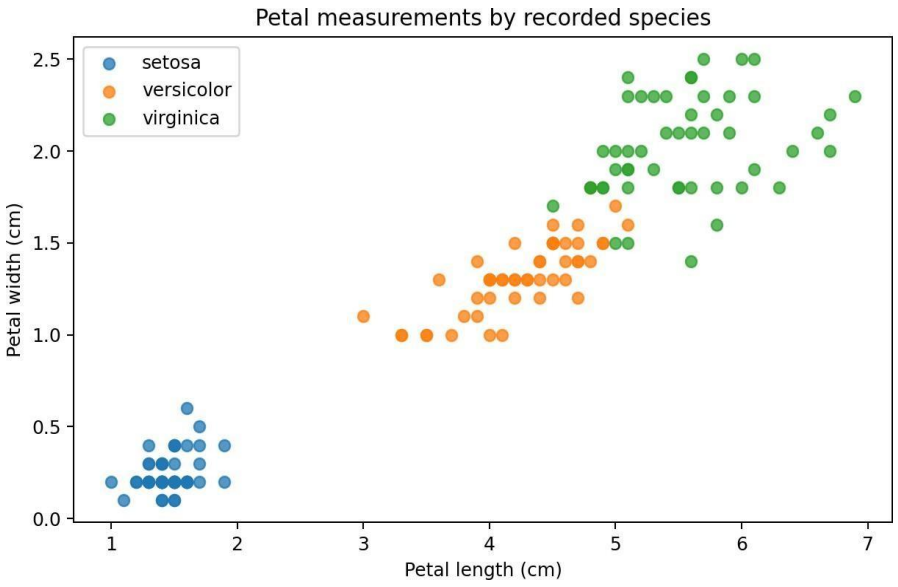


Figure 1. Petal length versus petal width by recorded species. Bias / measurement counterexample

If the flowers were collected from a narrow geography, cultivar set, or environmental range, the observed species separation may remain true for this sample while failing to describe other settings. The equal count of 50 observations per species is convenient for analysis but is not evidence that the species occur equally often in nature.

4. Prohibited claims

Do not claim: "Petal size causes an iris flower to become a particular species." The data are observational measurements of already-existing flowers and do not identify a causal intervention. Also do not claim that the 150 observations are representative of all iris flowers worldwide without additional sampling evidence.

5. Findings and limitations

The frozen dataset contains 150 observations, four numeric measurements, and three species labels. It has no missing values in this representation and contains one exact duplicate row. Petal measurements show strong sample-level separation, especially for setosa, but the conclusion is descriptive and bounded to this dataset.

Limited collection metadata constrain external validity and the duplicate row means row independence should not be assumed automatically.

6. Reproducibility and AI verification

The accompanying notebook is executable top-to-bottom, fixes a random seed, verifies the frozen file hash, performs schema/missingness/duplicate/range/anomaly/leakage checks, writes raw audit results, and saves the plot. AI was used to help structure and draft the work. The student should personally run the notebook, inspect outputs, and add any rejected AI advice or manual checks before submission.

Submission checklist

- Executed notebook

- This PDF report
- Data Card and data dictionary
- Frozen source data, license/source record, and SHA-256

- AI-use and independent-verification log
- 150-300 word result summary on the platform
- Stable GitHub repository URL with commit/tag